MA3104 2026-2027 Test 1

Time: 45 minutes Total marks: 35

1. Consider the differential equation:

$$(x+a)^2\frac{d^2y}{dx^2} - 3(x+a)\frac{dy}{dx} + 4y = (x+a)^2, \quad (x+a) > 0,$$

with $y(1-a) = 1$ and $\frac{dy}{dx}(1-a) = 0$, where y is dependent on x.

   (a) (4 points) Using a suitable transformation, express $\frac{dy}{dx}$ and $\frac{d^2y}{dx^2}$ in terms of derivatives with respect to z, the new independent variable. Transform the equation into a constant-coefficient differential equation in z.

   (b) (2 points) Transform the initial conditions into conditions in z.

   (c) (10 points) Solve the transformed equation and express the solution in terms of x.

2. Consider the system of differential equations:

$$\frac{dx}{dt} = 2x + y,$$
$$\frac{dy}{dt} = 2y,$$
$$\frac{dz}{dt} = 4z,$$

subject to the initial conditions $x(0) = 1$, $y(0) = 2$, $z(0) = 3$.

   (a) (5 points) Write the system in matrix form and find the eigenvalues of the coefficient matrix.

   (b) (2 points) Show that the repeated eigenvalue has only one linearly independent eigenvector.

   (c) (8 points) Find the general solution.

   (d) (4 points) Apply the initial conditions and give the final solution.